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Factors Contributing to Students Success on Algebra 1 End-of Course Exams

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Factors Contributing to Student Success on Algebra 1 End-of-Course Exams

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The thesis entitled '**Factors Contributing to Student Success on Algebra 1 End-of-Course Exams**' presented by **Mirranda A. Glasgow**, a candidate for the degree of **Master of Science in Mathematics,** has been approved and is worthy of acceptance.

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ABSTRACT

High school students in the state of Ohio are required to meet or exceed a specific score on several standardized tests in different subjects in addition to meeting coursework requirements to receive a high school diploma. These tests are called end-of-course (EOC) exams. As the name implies, these exams are given at the end of a required high school course and are based on the state standards for that course. Although these tests are aligned with the standards being used to create the curriculum, many students do not reach the required score when completing the test the first time. This prompted research to determine which factors, other than academics, are significant in predicting which students will not earn the score needed on the Algebra 1 EOC exam for graduation. Since these tests are so high stakes, teachers and administrators wish for students to earn the needed score on the first try. The purpose of this research is to determine which students may need remedial math courses to help prepare them to reach the desired score on the Algebra 1 EOC exam on the first try rather than needing to retake the test. The results of this study were obtained through a multiple regression analysis and two linear regression analyses. The multiple regression analysis examined Algebra 1 EOC exam scores as predicted by junior high math test scores, 8th-grade attendance, 9th-grade attendance, and free lunch status. This model was then reduced using a backward regression technique. This determined which factors were the most significant predictors of Algebra 1 EOC exam scores. The linear regression analyses were then conducted, choosing individual variables from those listed to have more insight into the predictors. The results of these analyses revealed that 9th-grade attendance and junior high math test scores are the most significant predictors of Algebra 1 EOC exam scores. It was also revealed that junior high math test score was a significant predictor of Algebra 1 EOC exam scores, and free lunch status was a significant predictor of 9th-grade attendance. The

results of this study imply that students with existing learning gaps, as evidenced by lower junior high math test scores, are likely to continue to have learning gaps and achieve lower test scores on the Algebra 1 EOC exam. The results also show that students with low SES are likely to miss more days of school than students with middle and high SES. Since attendance impacts a student's ability to score high on the Algebra 1 EOC exam, and SES impacts attendance, one can conclude that SES indirectly impacts a student's ability to score well on the exam. From these results, it can be inferred that students with poor attendance records and lower scores on prior standardized math tests will perform at lower levels on the Algebra 1 EOC exam and will likely need remediation prior to taking the exam.

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Introduction

Many states have recently been making a shift to high stakes End of Course (EOC) exams to determine student eligibility for high school graduation. As of 2019, Florida, Indiana, Louisiana, Maryland, Massachusetts, Mississippi, New Jersey, New Mexico, New York, Ohio, Texas, Virginia, and Washington include some form of these tests to be part of high school graduation requirements. (Gewertz, 2020). These tests are often the major determining factor in whether or not a student can graduate high school. Many students do well enough to pass these tests and continue their high school academic careers without this major roadblock. However, many students struggle to pass these tests on their first try for any number of reasons. This research plans to examine previous math test scores from the students' 8th-grade year, along with their free-lunch status, and attendance records to determine which of these factors greatly influences Algebra 1 EOC scores in 9th grade. This research will help many students who struggle with standardized testing to succeed by finding a way to provide interventions to alleviate obstacles to passing the test.

Background of the Problem

Standardized testing has long been evolving as a means to assess student progress and preparedness. In the beginning, it was mainly used to measure intelligence and college preparedness. Although created in the 1920s and 1930s, the SAT, ACT, and PSAT, became more widespread during the 1960s as a way to assess the college readiness of high school students, and as a benchmark for college acceptance. (NEA, 2020). In 2001, the No Child Left Behind (NCLB) Act mandated standardized testing for most grades in elementary and high school as a means to assess school effectiveness, student achievement, and teacher effectiveness. (NEA, 2020). In 2015, the Every Child Succeeds Act (ESSA) reformed standardized testing in the United States. This act lessened the volume of standardized testing required nationally, while still mandating high stakes testing for students in grades 3 through 8 for various subject areas. (NEA, 2020). Over the past century, standardized testing has evolved to be high stakes for students and schools, affecting many areas from grade level progression to high school graduation, and school funding. It has become increasingly important for students to achieve high scores on these tests to be considered competent in specific subject areas.

It is widely accepted by educators and administrators that students perform better on formative, summative, and standardized assessments when they attend school regularly. School attendance has been considered an important aspect of education since the early days of public schooling. In 1852, Massachusetts published the first law requiring parents to send their children to local schools a certain number of days per year. (Katz, 1976). As time progressed, more states passed laws requiring school attendance, and by 1918, all states currently in the union had passed at least one law requiring children to attend primary school. (Katz, 1976). Along with the advent of compulsory school attendance laws, came studies on the effects of the lack of school attendance. In a study conducted in 2012 by Parke and Kanyongo, it was determined that higher rates of absences were correlated with lower mean standardized test scores in mathematics. (Parke & Kanyongo, 2012). Another study conducted at the University of Tennessee found similar results. This study determined that the factor with the greatest influence on academic achievement was school attendance. Lower levels of attendance were associated with lower levels of academic achievement. (Hale, 2015). Based on these studies and many others like them, it is evident that attendance is an important factor in student success. As more emphasis is being placed on high-stakes testing and student achievement, attendance is more important than ever before.

Research Problem

In 2022, only 45% of students in the state of Ohio received a score of proficient, 700, or above on the Algebra 1 EOC exam. (ODE, 2022). This leaves 55% of students statewide who will need to retake the exam and possibly receive intervention services during or after the school day to help them earn a passing score on the exam. This extra intervention can often be time-consuming and frustrating to students who need it because they didn't pass the EOC exam. It takes away from the time that could be used to study for current courses or participate in extracurricular activities. Some factors may contribute to a student's success or unsuccess that are out of their control, including socioeconomic status and attendance. This study will explore how these factors contribute to student success on Algebra 1 EOC exams to be able to provide early intervention.

Purpose of the Study

Many students struggle with EOC exams, particularly with mathematics-based exams, Algebra 1, and Geometry. This study plans to examine factors contributing to success on the Algebra 1 exam because the Geometry exam does not factor into graduation requirements for students. If students do not receive a passing score of 684 after 2 attempts at any EOC exam, they must either continue to retest, or they can proceed to complete alternate pathways to graduation. These pathways include taking college credit plus (CCP) courses in the subject area, earning career or technical certifications, or enlisting in the military. (ODE, 2022) These alternate pathways may have their benefits, but they are not feasible for every student, and often require more time and commitment than just receiving a proficient score on the EOC exam. Oftentimes, students may not be receiving passing scores on these exams because of factors out of their control, such as their socioeconomic status (SES) and school attendance. This study will explore these factors' roles in student success on the Algebra 1 EOC exam. This research will help provide a way to identify students at risk of not reaching a proficient score when taking the test for the first time. Once factors contributing to student success are identified, as a school district, this information can be used to help schedule students in classes specifically tailored to help them do well on the Algebra EOC exam. Although the factors being identified may be non-academic factors, the factors are ultimately affecting students' academic progress. So, identifying the students before beginning their 9th-grade year, will allow the district to help students overcome the hurdles associated with these factors. This research may ultimately contribute to creating a new procedure for better preparing students for Algebra EOC exams and setting them up for future success in their high school academic careers.

Significance of the Study

This study is motivated by the need to help students succeed in achieving a proficient or competent score on Algebra 1 EOC exam scores as defined by the Ohio Department of Education. Helping students meet this goal will set them up for meeting the goal of graduating from high school. This exam is one of the main requirements to obtain a high school diploma in the state of Ohio, and many students struggle to achieve such a score of 684. If significant predictors of success on this exam are discovered, it could help schools place students in courses or other additional academic programs to set them up for success before taking the exam for the first time. Many students do not meet the required score when taking the exam as a freshman in high school. Then, they are required to take the exam again in the winter and/or spring of their sophomore year. Preparing for the exam is more difficult at this stage as many students are not enrolled in an Algebra class at this point, as they often pass the class, but not the EOC exam given by the state. Predicting a student's odds of success and placing them with additional help based on those odds may increase the likelihood that they will pass the test in one try.

Primary Research Questions

Question 1: Are students' socioeconomic status and level of attendance significant predictors of Algebra 1 End of Course Exam scores?

Question 2: Is there a significant relationship between prior standardized test scores in middle school math (6th, 7th, or 8th grade) and Algebra 1 End of Course Exam scores?

Research Design

This study will use data collected from approximately 220 students in grades 10 and 11 who have previously taken the Algebra 1 End of Course (EOC) exam. The number of absences during their 8th and 9th grade school years, prior math test scores from either 6th, 7th, or 8th grade, free-lunch status, and Algebra 1 EOC exam scores will be collected to examine which variables are significant predictors of students' EOC exam scores in Algebra 1. The major focus of the study will be on determining if the number of absences and free-lunch status (an indicator of socioeconomic status) are significant predictors of Algebra 1 EOC exam scores. A multiple regression analysis will be conducted using prior test scores, free-lunch status, and the number of absences as the independent variables, and Algebra 1 EOC exam scores as the dependent variable. This analysis will determine which variables are significant predictors of EOC exam scores. Additionally, a single variable regression analysis will be run to determine if there is a statistically significant relationship between prior math test scores and the Algebra 1 EOC exam scores.

Theoretical Framework

It is widely accepted that humans have psychological and physiological motivators for the things that they achieve in life. Maslow's Hierarchy of Needs theory expounds on this by asserting that humans are motivated by basic needs separated into different tiers and that one cannot move onto a different tier without first satisfying the needs from the previous tier. (McLeod, 2007). This theory was first published by Maslow in 1943 and included 5 tiers of needs that people strive to meet over a lifetime, including, physiological needs, safety needs, the need for belongingness, esteem needs, and the need for self-actualization. (McLeod, 2007). Just like any theory rooted in the human experience, Maslow's theory evolved and changed as time progressed. In the 1970s, Maslow added more levels to his hierarchy to better differentiate human needs. The version published in the 1970s included 8 tiers of needs. These 8 tiers are physiological needs, safety needs, need for belongingness, esteem needs, cognitive needs, aesthetic needs, the need for self-actualization, and the need to transcend and share actualization. (McLeod, 2007).

This theory has connections to education, many of which can be observed just by stepping inside a classroom for a few days. Just as many adults are continuing to meet their many tiers of needs, children, and adolescents are also working to meet and maintain their needs. This can often be more difficult for children as many do not have control over their environment. When students are not able to have their lower levels of needs fully met, they cannot meet higher levels of needs. As evidenced in Maslow's theory, esteem, and cognitive needs are higher up in the hierarchy as the fourth and fifth levels of needs. This means that a person's need for basic things such as food and water, their need for a safe environment, and their need to be accepted and have a feeling of belonging among peers and friends need to be met before they begin to strive for things that fall under the esteem and cognitive levels. This is where the desire for knowledge, learning, and achievement fall. (McLeod, 2007). Many students who live in poverty are not having their lower levels of needs met. This would make it more difficult for them to focus on higher need levels, which could lead to a lack of academic achievement. Which needs are being met, and to what extent have a great impact on a person's motivation to focus on learning things unrelated to meeting basic needs. This connects greatly to students experiencing poverty, as they are focusing on meeting basic needs with lower motivation to meet higher-level needs leading to lower levels of achievement in school and on standardized tests.

Assumptions, Limitations, and Scope

There are two main assumptions for the data collected for this study. The first assumption is related to standardized tests and EOC exams. This study assumes that the students strived to attain the best score possible on each of their standardized math tests, and the Algebra 1 EOC exam. The other assumption associated with this study is related to school attendance. This study assumes that students attended school as much as possible. There are limitations in this study related to test scores. The first is that data was missing for middle school math standardized test scores. Data was missing due to multiple factors. One group of students did not have 8th-grade math test scores because school was shut down due to the Covid-19 pandemic, thus canceling testing. 8th-grade math test scores were also missing for any students that completed Algebra 1 in 8th grade. In this case, 7th-grade math scores will be used. Finally, there was a group that did not have 7th or 8th-grade math scores. These were students that had their 7th-grade year interrupted by the Covid-19 pandemic school closings and completed Algebra 1 the following year during 8th grade. In this case, 6th-grade math scores will be used.

The scope of this study encompasses rural school districts. Since the data collected for this study is from a rural school district in Ohio, this study will be generalizable to similarly rural school districts in Ohio. This study is not necessarily generalizable to rural school districts in other states, because each state has standards and standardized testing unique to the state.

This study also has delimitations. The data collected for this study came from a single jr./sr. high school in a rural district in Ohio. The data will only be sourced from students currently in 10th and 11th grade, as 9th-grade students have yet to take the Algebra 1 EOC exam, and 12th-grade students did not take the Algebra 1 EOC exam due to Covid-19-related school closings.

Definitions of Terms

School Absence: When a student does not attend school for all or part of a day.

Standardized Test: A test given to all students of a specific grade and/ or subject area throughout a whole state or region.

Algebra 1 End of Course (EOC) exam: A test administered to all students after completing Algebra 1 in ninth grade, or the school year in which a student completes the course. (ODE, 2022)

Competency Score: To show competency on the Algebra 1 EOC exam must attain a score of 684 or higher. Reaching this score is a key component of Ohio's graduation requirements. (ODE, 2022)

Free-Lunch Status: A student may receive a free lunch if the average income in their household is below 200% of the poverty level specified based on the number of people living in the household. (benefits, n.d.) This status can be indicative of a student's socioeconomic status.

Summary

In summary, this study will focus on student achievement on the Algebra 1 EOC exams, with free-lunch status, attendance, and prior standardized test scores in math as the independent variables. A multiple regression analysis will be used to determine if free-lunch status and attendance are statistically significant predictors of Algebra 1 EOC exam scores. This study is centered around the idea that people need to have basic needs met before more abstract needs can be attained, and it hypothesizes that students with lower socioeconomic status and higher levels of absences will score lower on the end-of-course exam for Algebra 1.

The results of this study will be used to help future students reach the goal of attaining a competent score on the Algebra 1 EOC exam. If this study yields that free-lunch status, attendance, or both are significant predictors of Algebra 1 EOC exam scores, then this will be considered with upcoming students entering high school to determine if students should be placed in an intervention mathematics course to help boost math skills. This study aims to create a school environment that better provides students with the skills and resources needed to reach a competent score on the Algebra 1 EOC exam in one attempt rather than several.

Chapter 2

Introduction

This chapter will discuss several topics to provide background information on standardized testing in the United States. This chapter will discuss the following issues: the history of American education, the history of standardized testing in the United States, the prevalence of bias in standardized testing, Maslow's Hierarchy of Needs theory and its application to education, and chronic absenteeism in high school. The first two topics will provide background on the central issue of this study, student success with standardized testing. The latter three topics will provide information on matters that hinder student success on these tests. After reading this chapter, readers will better understand how standardized testing has become mainstream in American education and why it is a tremendous struggle for many students.

The History of Education in the United States

Education in the United States has not always been compulsory or accessible to all children. The first public schools in the United States existed mainly in the New England region of the country. These schools, often led by community leaders expected to exhibit strict morals, fulfilled needs that may not have been met in the home. (Paterson, 2021). The first example of compulsory schooling laws is also from the New England region of the United States, specifically, the area that would become Massachusetts in the 1640s. This law mainly focused on ensuring children received adequate religious and moral education. (Katz, 1976). Local members of the community often staffed these schools. (PBS, n.d.). However, since these schools were controlled by the small local communities they were built, they often needed to be more consistent. Many communities did not build schools even when required by law for various reasons. Thus, leaving children without the benefit of essential reading and math instruction. (Katz, 1976).

Beginning in the 1820s, common schools became the norm. This is the earliest version of public education in the United States. These schools enrolled students from all social classes and were funded by money from the community and fees paid by parents. (Goldin, 1999). These schools educated students in reading, writing, basic mathematics, history, geography, and grammar. The purpose was to provide children with enough education to become productive members of society. (Kober, 2020). Although children from all social classes were being enrolled, the education system was far from equally accessible. Students of color and students with disabilities were far less likely to be enrolled in a common school. (Kober, 2020.) Throughout the 19th century, the idea of educating all children of ages now typically associated with elementary grades took hold. By the turn of the century, most urban and rural children received the equivalent of an 8th-grade education. However, it was only at the end of the 20th century that nearly all children received a high school diploma. (Kober, 2020).

In the 1910s and 1920s, students began to more frequently enroll in high schools than previous generations, although completion rates were low compared with the modern day. (Kober, 2020). This increase in high school education led many academics to consider what should be taught in these higher-level schools. William Kilpatrick was one of these academics. He was a strong proponent of only teaching students subjects that would have a practical application to their adult lives. He suggested that students only enroll in upper-level math courses if those courses were of direct interest or use to the students. (Klein, 2003). By the 1930s, Kilpatrick’s philosophy had spread across all grades. Schools embraced the idea of determining courses based on student needs and interests rather than just teaching specific subjects. This led to integrating multiple subjects or topics into a single lesson. (Klein, 2003).

The 1930s style of education gave way to a significant deficit in basic math skills among young people. In the 1940s, this consequence became apparent and led to an increased emphasis on mathematical and other practical skills in high schools. This led to a decrease in more advanced math courses such as geometry and trigonometry. This movement ultimately fizzled out and was replaced with a curriculum that promoted an understanding of why mathematical procedures worked and how they worked. This movement was popular well into the 1960s (Klein, 2003). During the 1950s and 1960s, education faced the beginnings of what would prove to be a significant change in education, integration. (PBS, n.d.). During this era, schools were forced to integrate, leading to greater diversity in schools at all levels. This movement is often credited with creating better educational opportunities for students of color.

In the 1980s, national standards for education were in the beginning stages of implementation. During this decade, students began to have their educational success measured based on how well they met the specific standards. (Paterson, 2021). These standards were officially published in 1989. They emphasized the need for students to focus on problem-solving strategies in mathematics rather than computation. The National Council for Teachers of Mathematics (NCTM) promoted the use of technology to help students who lack basic skills to reach problem-solving skills. (Klein, 2003). The adoption of these standards in school districts across the nation presumably led to the adoption of national standards in the 1990s.

By the late 1990s, the majority of states had adopted academic standards in the areas of reading and math. (Paterson, 2021). The federal government enacted many national education reforms from the 1990s to the 2010s that gave way to the current educational environment. In

1994, President Clinton signed the Elementary and Secondary Education Act (ESEA) that provided additional funding for Title I programs in public schools, created higher standards for curriculum, and allowed for standardized testing to measure the attainment of the standards. (Goals 2000 and ESEA, n.d.). In 2001, President Bush signed the No Child Left Behind (NCLB) Act. This act aimed to improve math and reading skills in elementary and middle grades and provided funding for schools to encourage higher achievement of state and national standards. (Bush, 2001). In 2010, the Common Core State Standards (CCSS) were published. These standards were an attempt to create a set of national standards that would ensure all students in each of the 50 states were receiving the same level of instruction. (Kendall, 2011). In 2015, the Every Student Succeeds Act (ESSA) replaced NCLB and renewed the ESEA. This act pushed for the adoption of rigorous state standards for education. However, it did not force states to adopt the CCSS. It aimed to increase rigor and achievement for students at all levels. (Education Week, 2015).

In conclusion, education in the United States has been an ever-evolving institution. The many reforms have created the current educational environment. The reforms made in the late 1990s and 2000s led to an educational climate that heavily emphasized rigorous standards, especially in the areas of reading and math. Along with a strong emphasis on rigor came a focus on achievement, which would be measured almost exclusively using standardized tests in elementary, middle, and high school grades.

The History of Standardized Testing in the United States

Just as educational standards and practices have evolved, the methods of measuring the success of these practices have evolved. Over time assessments have been created to measure college readiness, career readiness, and teacher effectiveness. However, the main goal of these assessments has remained generally the same, to determine if students are learning the topics being taught at an acceptable level. Eventually, these tests evolved to have more significant consequences and determine the ability of a student to move on to the next grade or determine eligibility for high school graduation.

At the beginning of public education, school districts used formal assessments to determine how students progressed academically. The results of these assessments affected policy decisions for these small schools. (NEA, 2020). In the early 20th century, intelligence tests became a popular form of standardized testing. This testing was used to determine if students could succeed in public school. (Gershon, 2015). This standardized testing became increasingly popular throughout the early 20th century and became widespread by the end of World War I. (NEA, 2020).

During the same time that intelligence testing became widespread, college entrance exams were also becoming popular. In the early years of the 20th century, the College Entrance Examination Board is established. This board created tests in several subjects to be taken by prospective college students. However, these tests were still mainstream and often varied by the university. (NEA, 2020). These tests evolved during the first two decades of the 20th century. Then, by the early 1920s, the first version of the Scholastic Aptitude Test (SAT) was published and used for college entrance. (Gershon, 2015). This test also changed throughout the 1920s, and by the early 1930s, the SAT took on the form known today with a verbal and mathematical section. (NEA, 2020).

Another phenomenon that began during this time was using standardized tests in schools. These tests were typically used to measure academic achievement in high school students. By the 1930s, the Stanford Achievement Test sold millions of copies yearly. Around this time, Iowa became the first state to administer a statewide achievement test for high school students. (NEA, 2020). Similar exams began to show up in California and New York as well. All three states' exams aimed to measure a student's proficiency on specific topics. This was done to ensure a high academic ability since high school graduates were often guaranteed admittance into a state university. (Goldin, 1999). By the end of the 1930s, the Iowa Test of Educational Development was made available and adopted by several different states. This is just the beginning of mainstream standardized achievement testing. (NEA, 2020).

The following significant change in standardized testing practices occurred in the 1970s and 1980s. This was the beginning of the era of competency or proficiency testing. Between 1970 and 1980, nearly 75% of states had begun to require high school students to demonstrate a minimum competency level in specific subjects before graduation. (Dawson, 1985). In the state of Ohio in 1987, laws were passed requiring proficiency testing. The initial law required students of the graduation year 1994 to pass a proficiency test in math, citizenship, reading, and writing during their ninth-grade year to graduate from high school. And by 1996, students in fourth and sixth grades were taking proficiency tests in the same areas as ninth graders, with an added section in science. (ODE, 1998).

In 2001, the No Child Left Behind (NCLB) Act was signed. This act made statewide standardized testing mandatory in most elementary to high school grades. This act would also use the test scores as a means to evaluate the effectiveness of the schools. (NEA, 2020). The primary purpose was to use testing to measure how well students were being educated in reading and math. This was meant to help close the achievement gap. This act mandated that 4th and 8th-grade children take a national standardized test, while annual tests in other grades would be state-designed exams. (Bush, 2000). This act opened the gates for standardized testing that would become the norm.

The NCLB act prompted Ohio to create tests to measure student achievement more closely aligned with NCLB. In 2005, Ohio began administering these newly designed assessments. The achievement tests were administered in grades 3 through 8. These tests covered the topics of reading and math in all the previously mentioned grades, writing in 4th grade, and science and social studies in grades 5 and 8. These newly designed tests also included a test required for graduation. The Ohio Graduation Test (OGT) was administered to 10th-grade students, encompassing math, science, reading, writing, and social studies. As the name suggests, students were required to pass this assessment to graduate from high school in Ohio. (DeMaria, 2017). This would pave the way for even more high-stakes testing necessary for graduation.

In 2015, the Every Student Succeeds Act (ESSA) was signed into law. ESSA removed some of the stringent requirements of the NCLB Act but kept the need for standardized testing. Under ESSA, schools were required to administer reading and math tests annually to students in grades 3 through 8. Schools were also required to administer an assessment that students must pass to receive a high school diploma. ESSA allowed for several indicators to display student success. This meant testing was still an important indicator, but it was not the only indicator under this new act. (Education Week, 2015).

In 2015, Ohio changed its testing requirements for several grade levels. The testing requirements remained the same with an English Language Arts and Math assessment in grades 3 through 8. However, assessments for social studies were added in grades 4 and 6, and science assessments were added for grades 5 and 7. The graduation testing requirements also received a makeover. The OGT was no longer being used to determine graduation eligibility. Instead of the

OGT, students were required to complete End of Course (EOC) exams in several areas, including Algebra I, Geometry, English I, English II, American History, American Government, and Biology. These tests were required for graduation. (DeMaria, 2017).

Currently, ESSA continues to be the national program guiding state standardized testing requirements. However, testing requirements in Ohio have changed. One aspect of testing has remained constant; students must complete math and reading exams in grades 3 through 8. Students no longer complete social studies exams in elementary grades, and students now complete science exams in grades 5 and 8. (ODE, 2022). Graduation testing requirements have also changed slightly since 2015. Currently, high school students must complete EOC exams in Algebra I, Geometry, English II, American History, American Government, and Biology. However, students only need to reach a competency score of 684 on the English II and Algebra I exams. Students must only complete the EOC exams in all other areas; no minimum score is required for graduation. (ODE, 2022). This change has added increased emphasis on the areas of English and Algebra in high school.

Standardized testing has evolved on the national and state level throughout the last 100 years. It has become the standard instrument used to determine the effectiveness of teachers, schools, and curricula on a state and national scale. Standardized testing determines if a student has retained enough knowledge from prior coursework to be promoted to the next grade level and eventually deemed eligible for high school graduation. This places a greater emphasis than ever before on preparing students to achieve a specific score on each exam. However, there are many things out of the students', teachers', and school districts' control that can interfere with student scores.

Poverty and Student Achievement

It is well known that outside factors can affect a person's performance on physical or cognitive tasks in their everyday life. This is true of adults and children. However, children have even less control over their situation than adults. One such factor is the level of poverty in which a person lives. According to data from the National Center for Children in Poverty, about 38% of children are members of low-income families. Of this 38%, 17% fall below the federal poverty threshold, and 21% fall just above the federal poverty threshold. This means that approximately 2 of every five students live in a household where caregivers are likely struggling to make ends meet. (Koball et al., 2019). This shows that many students are struggling with the effects of poverty outside the school setting and are likely suffering secondary effects at school.

Parental employment is often a factor in whether or not a family experiences poverty. Although students living in poverty likely have one or more unemployed or underemployed caregivers, 57% of children in low-income homes have at least one caregiver working full-time. (Koball, et al., 2019). This implies that students have less home interaction with caregivers. With less time spent at home with the children, caregivers will likely be less involved in a student's education. Students with caregivers who are involved in their education are likely to have higher levels of academic achievement. (Chen, 2022). This places students from low-income families at a disadvantage.

Poverty has the potential to impact student achievement and success. Students who live in poverty are prone to emotional disregulation. This is related to students having a higher level of frustration when learning topics that are even the slightest bit challenging. These students often quit trying out of frustration. (Jensen, 2009). Impoverished students also likely lack social regulation skills. This makes it more difficult for these students to work cooperatively with their peers. The combination of poor social and emotional regulation set these students up for more significant academic struggles. (Jensen, 2009).

Stunted social and emotional regulation are not the only obstacles to students living in poverty. Children living in poverty often experience chronic stress. Chronic stress affects physical and psychological aspects of children's lives and can affect brain development and academic achievement. (Jensen, 2009). A study conducted in North Carolina found that low-income students were performing significantly lower on mathematics and reading standardized tests than their middle and higher-income peers. This study also found that schools with a higher percentage of low-income students performed worse on state reading and math assessments overall compared to schools with fewer low-income students. (Davis, 2021). In conclusion, students from low-income households are put at an academic disadvantage for many reasons.

There are many ways that living in poverty can set a child up with obstacles to their cognitive development and academic achievement. Studies have shown that children in poverty have a slower rate of vocabulary acquisition in the early years of speech and language development. (Jensen, 2009). Reading is also often affected in low-income children. Since many children from low-income families do not have a caregiver at home full-time, these children have fewer opportunities to have adults read to them and engage in conversation with adults. This affects their ability to build vocabulary, phonemic awareness, fluency, and comprehension. (Jensen, 2009). This tends to put low-income students at a disadvantage from the very beginning. Students from low-income homes typically perform more poorly on assessments, which leads to a lowering of expectations from teachers and other educators. (Jensen, 2009). Having low expectations for a student often gives them a reason to perform at a lower level than their peers. This may lead to consistently poor academic performance.

Overall, it can be seen that poverty typically has adverse effects on academic achievement in all grade levels, but especially elementary grades. (Jensen, 2009). Once a student develops a skill deficit in early grades, it is difficult to make up for that deficit in later years. This puts them at risk for low academic achievement. Poverty often creates a wealth of issues, including low self-esteem, absenteeism, and lower cognitive functioning, that lead to low academic achievement. (Jensen, 2009).

Absenteeism and Academic Achievement

Absenteeism is a problem that many schools in the United States face. When students are absent from school, they miss instructional time in each of their classes. When students are chronically absent from school, they miss several hours of instructional time in each content area each week. This type of absence can put a student behind their peers academically. (Balfanz & Byrnes, 2012).

There are many reasons a student may be absent from school on a regular basis. Students may miss school because they are struggling academically and developing low confidence in their ability to do well in school. (NAESP, 2016). Students may be getting bullied at school, and being absent from school is their way of avoiding the bully. (NAESP, 2016). These are both issues that can snowball into much bigger attendance issues and should be addressed by school professionals.

Students may also suffer from health issues that make them miss multiple days of school. (NAESP, 2016). This is something that can be temporary, or the student may be suffering from a chronic illness that causes them to miss many days of school. In addition to physical health, students may miss school days for mental health needs. Students may suffer from a mental illness that prevents them from functioning well at school. (NAESP, 2016). Physical and mental health issues may cause even more absences for low-income students, as they are more likely to suffer from health issues. (Jensen, 2009). Students living in poverty often have inadequate living conditions, which can cause respiratory illnesses and physical injuries. These children may also lack health insurance coverage which can cause them to suffer from minor illnesses for a longer period of time than middle and high-income peers. (Jensen, 2009). Low-income students are also more likely than their peers to develop depression during adolescence, contributing to more absences from school. (Jensen, 2009).

Students may also miss school because they are helping to care for a member of their family that has fallen ill or otherwise requires extra care. (NAESP, 2016). This is common in students from low-income families as they are often the caregivers for younger siblings when their parents are not around. (Jensen, 2009). This situation is common yet unfortunate. Many adults who are working long hours or multiple jobs to ensure bills are paid rely on older siblings to care for younger siblings, even during times of sickness. (Jensen, 2009).

Students may be absent from school because their family is struggling to maintain consistent housing. (NAESP, 2016). Impoverished students may experience transience for any number of reasons related to their socioeconomic status. These moves to different places are often involuntary on the part of the family. (Jensen, 2009). Frequent moves could cause a child to change schools or even school districts multiple times within a school year, which can be disruptive to learning.

Parke and Kanyongo (2012) studied the prevalence and impact of attendance, or lack of attendance and mobility, on students moving to different schools within or outside the district. Students were observed in grades 1 - 12 in a Pittsburgh, PA, school district to determine rates of absence, mobility, and academic achievement. The researchers were specifically targeting mathematics achievement in grades 8 and 11. It was found that attendance had a significant effect on mathematics achievement in 8th and 11th grade when measured by standardized tests administered by the state of Pennsylvania. When controlling for SES and gender, students with higher rates of absences had lower mean test scores compared to students with more stable attendance records. When specifically analyzing high school students, it was determined that 7 of the ten high schools studied had shown a statistically significant relationship between rates of attendance/ mobility and standardized mathematics test scores. (Parke & Kanyongo, 2012). Their findings show how important school attendance is for student achievement, especially in mathematics, as mathematics is a subject that is constantly building upon previous content knowledge to be successful with a new skill or topic. If a student misses multiple days of school in a month, they are not being exposed to the skills enough to reach mastery level. Then they are asked to apply that prior knowledge to a new skill. This seemingly would create a cycle of learning gaps and lapses, leading to a student being unsuccessful in mathematics overall.

Hale (2015) studied whether low income, chronic absenteeism, and living in a one-parent household contributed to lower-than-average academic performance in high school students. Students in grade 12 at a high school in Chattanooga, TN were grouped based on whether they fit one or multiple criteria mentioned above. Those students' GPAs and standardized test scores were then analyzed to determine if there was a relationship between the factors mentioned and academic achievement. After analyzing the data, it was found that students who had missed more than nine days of school had a lower than average score on standardized tests and a lower-than-average GPA. It was also found that absenteeism combined with low income, a one-parent household, or both resulted in even lower levels of academic achievement than just with low income or a single-parent household alone. It was also determined that low SES and single-parent households were the only statistically significant factors of low academic achievement when combined with having a high number of absences. (Hale, 2015). This study shows again that school attendance plays a huge role in academic success. Although this study covered more than just achievement in mathematics, it looked at overall GPA, Biology, English, and Algebra scores. It has shown, just as the previously mentioned study, that missing several days' worth of information creates a cycle in which students cannot reach a mastery level of knowledge on a topic before being asked to use that prior knowledge and apply it to a new topic.

Chronic absences have consequences that may last for a student's entire academic career. Students who are chronically absent in kindergarten typically show lower performance than peers in first grade. This creates more of a disadvantage for students from low-income families, as those students may already have a skill deficit that is then compounded by missing school. (Balfanz & Byrnes, 2012). Chronic absences from school can create a learning gap at any age or grade level. (Balfanz & Byrnes, 2012). A learning gap, if not recovered, can lead to greater gaps in skills later in a student's academic career. There has been data to show that poor school attendance as early as 6th grade can affect when and if a student graduates from high school. (Balfanz & Byrnes, 2012). Students from low-income families are more likely to chronically miss school. In impoverished rural areas, approximately 25% of students are typically absent for a month's worth of school days. (Balfanz & Byrnes, 2012). This is often compounded over multiple years for many students in poverty. Many of these students accumulate six months to 1 year's worth of missed school days over five years. (Balfanz & Byrnes, 2012). These absences, especially in economically poor areas, are often linked to obstacles created by poverty. Since education is the most effective way out of poverty, this creates a vicious cycle that is difficult to break for many students who are not having their needs met. (Balfanz & Byrnes, 2012).

Maslow’s Hierarchy of Needs

Maslow’s Hierarchy of Needs Theory is a theory centered around human motivation. The original theory is comprised of a five-tier model showcasing physical and emotional needs that motivate human behavior. Two newer models with seven and eight tiers were introduced two decades after his original theory was published. (McLeod, 2018). According to this theory, lower-level needs must be satisfied before higher-level needs are satisfied. (McLeod, 2018). For the purposes of this study, the newer model with more tiers will be considered.

The expanded version of Maslow’s theory includes eight tiers of needs. The first tier includes the most basic of human needs, including food, shelter, warmth, and sleep. These are the most basic needs a human must satisfy to survive. (McLeod, 2018). The second tier is centered around safety. It includes security, stability, and order. If these needs are met, it gives a person a sense of security in their overall life. (McLeod, 2018). The third tier focuses on love and belonging. The needs in this tier include friendship, acceptance, and belonging to a friendly peer group. This fulfills the basic social and emotional needs of a person. (McLeod, 2018). The fourth tier focuses on esteem needs. This tier asserts that humans have a need for achievement, independence, and respect from others. (McLeod, 2018). These needs are the basic building blocks of self-confidence. The fifth tier in this model is all about cognitive needs. These needs include the need for knowledge, understanding, and finding meaning in things. (McLeod, 2018). The sixth tier is the need for aesthetics. This level includes the need for balance and appreciation of things humans find beautiful. (McLeod, 2018). The seventh tier is centered around self-actualization. These needs are more abstract and include realizing one’s potential and seeking personal growth. (McLeod, 2018). Finally, the eighth tier is transcendence. These are needs that extend outside oneself, including serving others, religious and spiritual needs, and scientific discovery. (McLeod, 2018). As it is apparent, these tiers of needs build upon each other like brick layers; without the first, you cannot have the second, and so on.

As the theory asserts, higher levels of needs cannot be met without lowers levels first being met. Students cannot fulfill social, esteem, or cognitive needs without first having basic psychological, physical, and relational needs met. (McLeod, 2018). This theory can be observed in public schools. Students cannot focus on gaining knowledge if they are not having their basic needs met at home. (Burleson & Thoron, 2014). In many ways, schools can help students meet some of their lower-level needs to facilitate learning. Unfortunately, schools cannot meet the most basic of needs, such as providing appropriate clothing, shelter, and sleep for students. However, educators can point children and parents to programs that will facilitate the acquisition of basic needs. (Burleson & Thoron, 2014). If students can get even a fraction of these needs met, it will allow them to focus more energy on learning. Although school faculty members cannot account for the basic survival needs of a student, they can account for safety needs. Teachers and administrators have the power to create environments within a school that allows students to feel safe. The existence of specific rules and procedures that are consistent and predictable creates a sense of stability and safety for students. (Burleson & Thoron, 2014). The final needs that educators are able to meet within the school are the needs for belonging and self-esteem. Fostering a positive learning environment allows students to have more positive social interactions with peers and adults. (Burleson & Thoron, 2014). This creates an environment that facilitates the formation of friendships and positive social relationships. Teachers can create bonds with their students by simply taking an interest in their individual needs and interests at school. By doing this, it shows the students that the teacher cares about their well-being. This small action may help fulfill the need for love and belonging that a student may be missing at home. (Burleson & Thoron, 2014).

Teachers and school administrators, and other faculty members may not have the power to attend to every lacking need that a student experiences, but they can create an environment that is welcoming and free from judgment or ridicule. This will allow students to ask for help, build relationships, and form trusting bonds. These steps alone can alleviate stressors from a student and allow them to focus more energy on cognitive needs such as learning.

Summary

To summarize, this chapter reported on a collection of literature and background information to further reinforce the purpose of the study. The history of education and standardized testing in the United States was discussed to give background on the relevance of the material being studied. This information made it clear that high-stakes testing and standards of learning have both risen in recent decades. Absenteeism, poverty, and academic achievement were discussed in relation to each other. The discussion of these topics revealed that poverty and absenteeism contribute to lower levels of academic achievement. Finally, Maslow's Hierarchy of Needs Theory was discussed in connection to education. This was used to show that students are less motivated to learn when their basic needs are not being met at home.

Chapter 3

Introduction

This chapter will discuss the methodology of this study. This chapter will provide a blueprint for how the researcher will analyze the data from this study and background information on the data collected. More specifically, this chapter will cover the study's setting, the participants' demographics, the instruments used to generate the data, the procedure for collecting such data, and the procedure for processing the data.

Setting and Participants

This study takes place in a rural area of the eastern Appalachian region of Ohio. The school district is located in a county with a population of 64,330, with 89.8% of the population identifying as white, 5.5% identifying as Black, 1.8% identifying as Hispanic, 0.2% identifying as Native American, 0.6% identifying as Asian and 2.5% identifying as multiracial. (U.S. Census Bureau, 2022). The area in which the district is located is known to be economically disadvantaged, with a median income of $49,211 and 17.2% of people living in poverty. (U.S. Census Bureau, 2022).

The high school at which the researcher collected the data follows the pattern set by the county for racial demographics. The majority of students are white, with a small minority of students identifying as Black, Asian, or Hispanic. The district is known to have a large population of economically disadvantaged students.

The participants of this study are high school sophomores and juniors that have completed an Algebra 1 course and have previously taken the Algebra 1 End of Course (EOC) exam in Ohio. These students would have also previously taken at least one mathematics exam in grades 6, 7, or 8. Due to the Covid-19 pandemic, schools were shut down in March 2020. This led to the disruption of schooling and testing for many students across Ohio, including this district. This means several students had at least one year of not completing a standardized mathematics test. This led the researcher to use test data from any previous mathematics test taken during 6th, 7th, or 8th grade.

For this study, the researcher chose two primary research questions; "Are students' socioeconomic status and level of attendance significant predictors of Algebra 1 End of Course Exam scores?" and "Is there a significant relationship between prior standardized test scores in middle school math (6th, 7th, or 8th grade) and Algebra 1 End of Course Exam scores?" The researcher has chosen to use multiple regression techniques to examine both of these questions. The examination of these questions requires a total of four predictor variables, socioeconomic status, 8th-grade attendance, 9th-grade attendance, and prior math scores. G*Power was used to calculate the suggested sample size for this analysis. When using a standard alpha of 0.05, a standard effect size of 0.15, and a standard confidence interval of 0.95, it is suggested to have a minimum sample size of 74.

Instrumentation

The Algebra 1 End of Course (EOC) exam and the standardized mathematics tests given in 6th, 7th, and 8th grades fall under the category of Ohio's State Tests (OSTs). The Ohio Department of Education (ODE) develops these tests with Cambium Assessment. Test questions are created by Cambium Assessment, based on Ohio's learning standards. Test blueprints provided by ODE determine the amount of each type of test question. These questions are then submitted to assessment specialists at ODE to be revised. After revision, these questions are then sent to various committees comprised of educators, community members, and parents to evaluate the questions' content, bias, and possible moral implications. College experts also review the Algebra 1 EOC exam questions to assess college readiness measures. One of these committees also creates rubrics to score any open-ended questions in OSTs. All the developed questions are then field tested on OSTs given to students without affecting students' scores. The field tests of these questions are then evaluated by ODE and Cambium to check the quality of the test items. Once complete, the items are either selected for the official test bank or returned for more revision and development. (ODE, n.d.).

Descriptive statics for each OST is listed in the *Operational Summary Statistics* table published by ODE each year. According to this document, the grade 6 math test has a reliability score of 0.94, the grade 7 math test has a reliability score of 0.92, the grade 8 math test has a reliability score of 0.91, and the Algebra 1 EOC exam has a reliability score of 0.92. (Office of Assessment, n.d.).

Each OST is based on a test blueprint created by ODE. These blueprints are published by ODE to provide insight to parents and educators about each test and its content. The tables below show how these math tests are segmented by topic.

Table 1: Grade 6 Math Test Raw Score Blueprint

Area of Mathematics	Ratios and Proportions	Expressions and Equations	Geometry and Statistics	The Number System	Modeling and Reasoning	Total Points
Amount of Raw Score Points	13 - 17	17 - 23	11 - 13	11 - 13	> 11 (Combined within other areas)	52 - 54

(Grade 6 Math Test Specifications, Ohio Department of Education, n.d.)

Table 2: Grade 7 Math Test Raw Score Blueprint

Area of Mathematics	Ratios and Proportions	The Number System	Geometry	Statistics and Probability	Modeling and Reasoning	Total Points
Amount of Raw Score Points	12 - 16	15 - 19	11 - 13	12 - 15	> 11 (Combined within other areas)	52 - 54

(Grade 7 Math Test Specifications, Ohio Department of Education, n.d.)

Table 3: Grade 8 Math Test Raw Score Blueprint

Area of Mathematics	Expressions and Equations	Functions	Geometry	The Number System	Modeling and Reasoning	Total Points
Amount of Raw Score Points	11 - 15	11 - 15	15 - 19	11 - 13	> 11 (Combined within other areas)	52 - 54

(Grade 8 Math Test Specifications, Ohio Department of Education, n.d.)

Table 4: Algebra Test Raw Score Blueprint

Area of Mathematics	Number, Quantities, Expressions, and Equations	Functions	Statistics	Modeling and Reasoning	Total Points
Amount of Raw Score Points	19 - 22	23 - 27	10 - 12	> 11 (Combined within other areas)	54 - 56

(Algebra Test Specifications, Ohio Department of Education, n.d.)

Each OST is segmented into four performance levels. These levels are Basic, Proficient, Accomplished, and Advanced. These levels indicate where students perform academically. The table below shows the score cut points for each performance level.

Table 5: Scaled Score Cutpoints for Performance Levels

Test	Basic	Proficient	Accomplished	Advanced
Grade 6 Math	682	700	725	744
Grade 7 Math	684	700	725	755
Grade 8 Math	690	700	725	744
Algebra 1	682	700	725	754

(OST Spring 2022 Statistical Summary, Office of Assessment, Ohio Department of Education, n.d.)

In summary, these tests measure similar mathematical concepts across different grade levels. The number of questions on each assessment is similar and provides a consistent tool to measure academic growth in mathematics over several years.

Procedure

The standardized test scores used for this study were tabulated before the beginning of this study. ODE tabulated the Algebra 1 scores in 2021 and 2022. ODE tabulated the 6th, 7th, and 8th-grade scores in either 2021, 2019, or 2018. No tests were given in 2020 due to the Covid-19 pandemic. The scores are calculated and collected by the Ohio Department of Education, and all teachers and administrators at the school district have access to the scores of all students.

The attendance records and free-lunch status of students are accessible to all teachers in each respective school building. These records can be accessed at any time by teachers through the school district's classroom management software, ProgressBook, which Omeresa manages.

The data used for this study is accessible to all teachers in the school district at any time. The data has also been cleansed of all identifying information. This means that there is not any added risk to students. The risk is no greater than on a typical school day. Since there is no added risk to students, parental permission was not required to perform this study.

The Institutional Review Board of Shawnee State University approved the exempt application for this study on November 14, 2022. This study received exempt status since it did not pose any additional risks to the confidentiality of students. This document of approval can be viewed in Appendix A.

Data Processing and Analysis

The data were collected and organized into a spreadsheet. The data was then cleansed of identifying information, and the subjects were labeled with arbitrary ordinal numbers. Any subjects with missing pieces of data were removed from the data set. The data collected is separated into six categories; the number of 8th-grade absences, 9th-grade absences, free-lunch status, Algebra 1 EOC exam score, junior high math OST score, and grade level. (Grade level indicates which grade students were in when they took their last recorded math OST before taking Algebra 1).

As stated previously in this chapter, the primary research questions for this study are; "Are students' socioeconomic status and level of attendance significant predictors of Algebra 1 End of Course Exam scores?" and "Is there a significant relationship between prior standardized test scores in middle school math (6th, 7th, or 8th grade) and Algebra 1 End of Course Exam scores?” A multiple regression analysis will be conducted to address both of these questions. This analysis will be conducted using the statistical software R version 4.2.2. The initial analysis will be conducted with 8th-grade attendance, 9th-grade attendance, socioeconomic status, and prior math OST scores as predictor variables. The following assumptions will be checked before proceeding with the analysis, normality, multicollinearity, and independence. The researcher will check the data for normality by examining a Q-Q Plot of the data set, and by performing a Shapiro-Wilk test. The possibility of multicollinearity will be explored by computing the variance inflation factors of the data. Finally, independence will be tested by plotting residual values over fitted values and looking for patterns in the data.

Once the assumptions are tested, the regression analysis will be run. The regression analysis will be initially run with all previously mentioned variables in place. The process will then be repeated, and one variable will be removed to check for significant output changes. This will be repeated until the only predictor variable left is prior math OST scores. This will help to determine which variables are significant predictors of Algebra 1 EOC exam scores.

The second research question will be addressed similarly. A multiple regression analysis will be conducted using the predictor variables listed in the previous paragraph. Then, an additional analysis will be run with the prior math OST score variable removed. This will show whether or not prior math OST scores are a significant predictor of Algebra 1 EOC exam scores.

The researcher is also interested in determining if the grade level during which the students completed their most recent math OST has a significant influence on whether the mast OST score is a significant predictor. So, an additional regression analysis with grade level and prior math scores as predictors will be run. Then an analysis with only prior math OST scores will be run. The results will be compared.

This technique has been used in other, similar studies. A study by Daffner, DuPaul, Anastopoulos, and Weyandt in 2022, researched the predictors of academic success in students with Attention Deficit/ Hyperactivity Disorder (ADHD). This study examined which pre-college and college demographics were significant predictors of academic success for these students as college freshmen. The independent variables examined were gender, race, socioeconomic status, past academic achievement, and cognitive skills. The researchers then used a multiple regression analysis to examine one of their primary research questions, which aimed to determine significant predictors of first-year college GPAs of students with ADHD. (Daffner, et al., 2022).

Another study using this technique was conducted by W. Johnson, A. Johnson, and J. Johnson in 2012. This study focused on predicting success on the Texas Chemistry STAAR test. This study looked at two predictor variables for success on the STAAR test. These variables were the grade in which the student took chemistry, sophomore or junior, and previous scores on the TAKS test given each year in science. A logistic regression analysis was used to accurately predict which students would succeed on the STAAR chemistry test, and which students would not succeed on the test. (W. Johnson, A. Johnson, and J. Johnson, 2012).

Summary

In summary, this study will focus on predicting Algebra 1 EOC exam scores from 8th and 9th-grade attendance, SES, and previous math OST scores. This will be accomplished using multiple regression analysis to determine which independent variables are significant predictors. The data being used for this study is anonymous and poses no threat of breaching the confidentiality of the participants. It required no additional risk to the students. The tests used as predictor variables are similar in skills to the dependent variable test. This increases the likelihood that the OST math scores will significantly predict Algebra 1 EOC exam scores. This technique has been used in other similar studies with successful outcomes.

Chapter 4

Introduction

Chapter 4 will contain the detailed results from data analyses conducted with a single data set. The analyses conducted included a multiple regression analysis and a linear regression analysis. This study was conducted to determine if rates of absence in school and socioeconomic status (SES), determined by free lunch status and Junior High math standardized test scores, are significant predictors of Algebra 1 End of Course (EOC) exam scores. By conducting these analyses, the researcher hopes to help students and teachers by providing a way to predict which students may need more assistance during the year in which they complete the Algebra 1 course. By providing a measure to predict likely success or failure on the EOC exam, students can receive any extra instruction needed to achieve the desired scores on the Algebra 1 EOC exam.

The data used in these analyses was collected from a single high school in rural Ohio. The data was collected on students from various grade levels who had already completed Algebra 1 and the Algebra 1 EOC exam. The data collected and recorded includes the following, Algebra 1 EOC exam scores, scores from one standardized math test taken during junior high, the free lunch status of each student, the number of days absent during 8th grade, and the number of days absent during 9th grade. The junior high standardized test scores are varied by the grade level in which students had taken the test. Since the Covid-19 pandemic forced schools to close and interrupted standardized testing during the 2019 - 2020 school year, some students did not have a standardized math score from 8th grade. Therefore, 7th-grade and even 6th-grade scores were collected to predict Algebra 1 EOC exam scores.

The following sections of this chapter will discuss the results of a multiple regression analysis and a linear regression analysis performed using the abovementioned variables. Both regression analyses were performed to determine which variables were significant predictors of Algebra 1 EOC exam scores. The results of the analyses will be discussed, along with descriptive statistics and demographics.

Participants

This study includes data from 213 participants. These participants are all high school students who have completed at least one standardized math test in junior high school and the Algebra 1 EOC exam. These students all attend school in the same district. Of these 213 participants, 119 receive a free or reduced-price lunch, and 94 pay full-price for lunch. This means that 119 students in this sample live in homes below the poverty line for income. Also, 147 students met or exceeded the competency score required on the Algebra 1 EOC exam. This means that 66 students did not meet the minimum competency score.

The means, standard deviations, range, and confidence intervals of each variable used in this study are listed below in Table 6. The variables are as follows: Number of absences in 9th grade (9th_Abs), number of absences in 8th grade (8th_Abs), free lunch status (free_lunch), Algebra 1 EOC exam score (alg_score), and standardized math test scores from junior high (jr_high_score).

Table 6: Means and Standard Deviations of Variables

	Mean	Standard Deviation	Range	95% CI
9th_Abs	20.815	16.224	(0, 99.86)	(-0.36, 0.01)
8th_Abs	12.749	11.564	(0, 86.82)	(-0.14, 0.38)
free_lunch	0.441	0.498	(0, 1)	(-3.98, 6.12)
alg_score	697.024	29.111	(618, 754)	(-19.21, 131.28)
jr_high_score	708.329	25.079	(633, 794)	(0.80, 1.01)

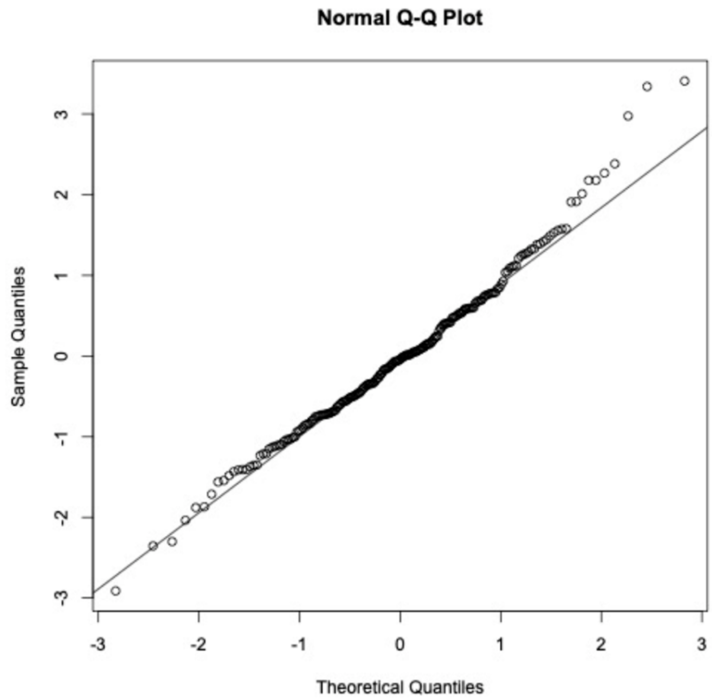
Data Analysis

This section will discuss the results of the regression analyses conducted with the data for each research question. The results will be broken down by question below. The descriptive statistics related to each question and analysis will be discussed concerning the conclusion for each question.

Question 1: Are students' socioeconomic status and level of attendance significant predictors of Algebra 1 End of Course Exam scores?

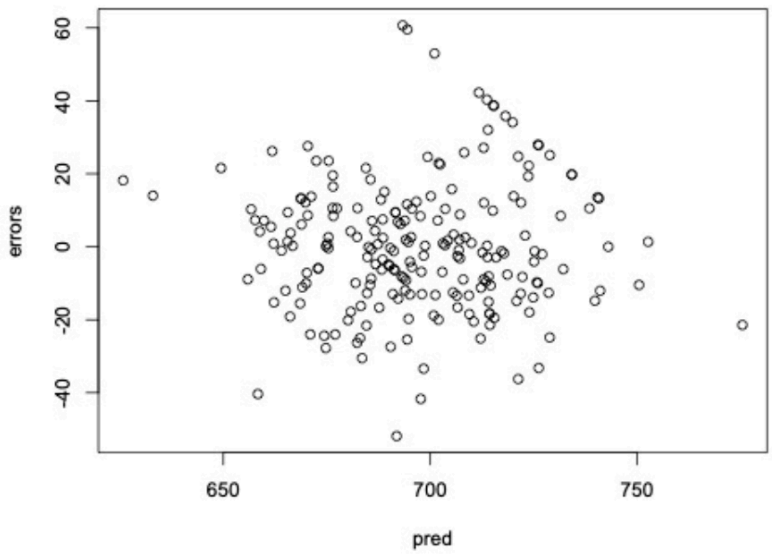
A multiple regression analysis was used to examine this question. Before conducting the analysis, several assumptions were tested to determine if the data was fit for this analysis. Variance inflation factors (VIF) were calculated for each predictor variable. The variance inflation factors are as follows; jr_high_score = 1.16, 9th_Abs = 1.49, 8th_Abs = 1.54, and free_lunch = 1.07. Since all of the VIFs fall into the range (1, 2), there is no evidence of multicollinearity that needs to be corrected. The diagonal values of the hat matrix were also examined. The range of these values is (0.009, 0.26). This range exceeds the threshold of 0.47, indicating that there are outlier values influencing the model. The normality assumption was examined by analyzing a QQ plot. This plot is shown below in Graph 1.

Graph 1. Normal QQ Plot



Upon examination of this graph, there appear to be some extreme values at the higher end of the scale. However, the majority of values fall on or near the line, indicating that the data being used comes from a mostly normal distribution. After examining the normality assumption, the assumption that all variances are equal was examined. This was examined using a graph of fitted values over residual values. Graph 2 shows the plot of fitted values over residual values.

Graph 2. Residuals vs. Predicted Values



Upon examination of the graph above, it can be determined that equal variances are not of concern in this data set. The data is spread out and does not appear to follow any pattern or form a cluster.

A multiple regression model was created to predict Algebra 1 EOC exam scores from 8th-grade attendance, 9th-grade attendance, free lunch status, and junior high standardized math test scores. The AIC of the full model, including four variables, is 1840.918. This value is relatively large, indicating that the model is possibly a good predictor of Algebra 1 EOC exam scores. The adjusted R^2 = 0.62. This indicates that 62% of the variation in Algebra 1 EOC exam scores can be explained by the 4 variables mentioned above. Table 7 below shows the estimated regression coefficients, standard errors, 95% confidence intervals (CI), and descriptive statistics for this multiple regression model. Table 8 shows the ANOVA table for the full multiple regression model.

Table 7: Descriptive Statistics for Full Multiple Regression Model

Dependent Variable	Independent Variable	Estimated Regression Coefficient	Standard Error	t-value	p-value	95% CI
Alg_score	**jr_high_score**	0.91	0.05	17.18	p < .001	(0.80, 1.01)
Alg_score	**9th_Abs**	-0.17	0.09	-1.88	0.06	(-0.36, 0.01)
Alg_score	**8th_Abs**	0.12	0.13	0.92	0.36	(-0.14, 0.38)
Alg_score	**free_lunch**	1.07	2.56	0.42	0.68	(-3.98, 6.12)

Table 8: Summary ANOVA Table for Full Multiple Regression Model

	Df	SS	MS	F value	p value
Regression	4	112737	28184.25	87.79	p < .001
Residuals	208	66828	321.29	----	----
Total	212	179565	----	----	----

After conducting the ANOVA of the multiple regression model, it was concluded that the full multiple regression model is statistically significant; $F(4, 208) = 87.79$ and $p < .05$. Since the model is statistically significant, it can be concluded that the multiple regression model using all four variables is a good predictor of Algebra 1 EOC exam scores. Upon examining this analysis further, it was revealed that the only statistically significant predictor in this model is junior high standardized math test scores $F(1, 208) = 347.6$ and $p < .05$. This shows that the model may benefit from having other predictors removed from the model.

After analyzing the full multiple regression model, the decision was made to create a reduced model using backward elimination. Creating a reduced model with backward elimination helped determine which variables were the most significant predictors of Algebra 1 EOC exam scores. The backward elimination model retained two predictor variables, 9th-grade absences and junior high standardized math test scores. This new model is a statistically significant predictor of Algebra 1 EOC exam scores with $F(2, 210) = 175.8$ and $p < .05$. The AIC of the backward model is 1838.03. This is less than the AIC of the full multiple regression model, indicating that the reduced model is a better predictor of Algebra 1 EOC exam scores than the full model.

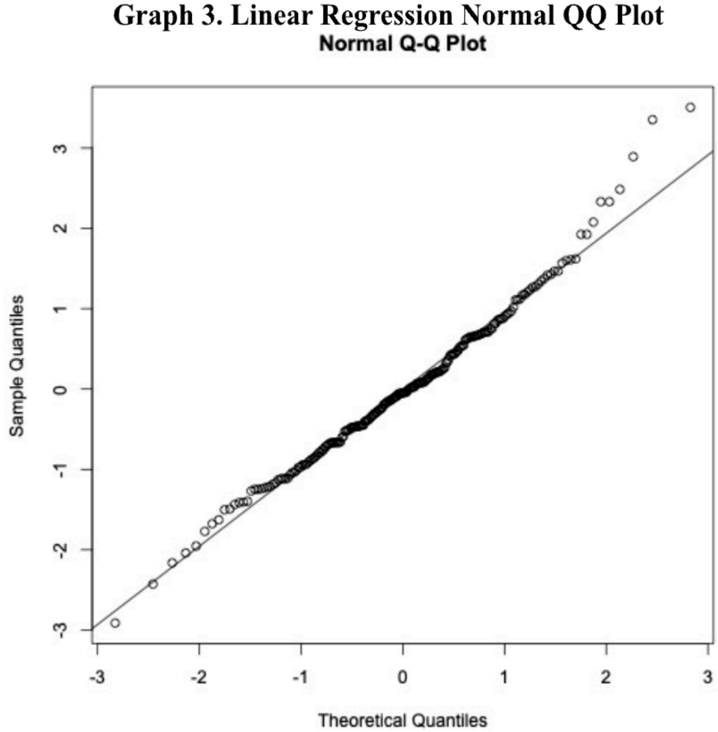
The results from the reduced model prompted the researcher to use linear regression to determine if there was a significant relationship between 9th-grade attendance and SES since 9th-grade attendance was a significant predictor of Algebra 1 EOC exam scores. A linear regression model was created with 9th-grade attendance being predicted by free lunch status. This model showed that free lunch status was a statistically significant predictor of 9th-grade attendance with $F(1, 211) = 9.66$ and $p < .01$. The estimated regression coefficient of this model is 6.81. This indicates that students with low SES are likely to miss 6.81 more days of school per year than students with middle or high SES. This indicates that although SES is not a significant predictor of Algebra 1 EOC, it is a significant predictor of 9th-grade attendance, which is significant in predicting Algebra 1 EOC exam scores.

In conclusion, the results of the multiple regression analysis have revealed that 9th-grade absences are a significant predictor of Algebra 1 EOC exam scores. This is evidenced by the backward regression model, including 9th-grade absences. This shows that an increased number of absences will likely lead to lower Algebra 1 EOC exam scores. This factor should be examined more closely by school administration to help alleviate low test scores.

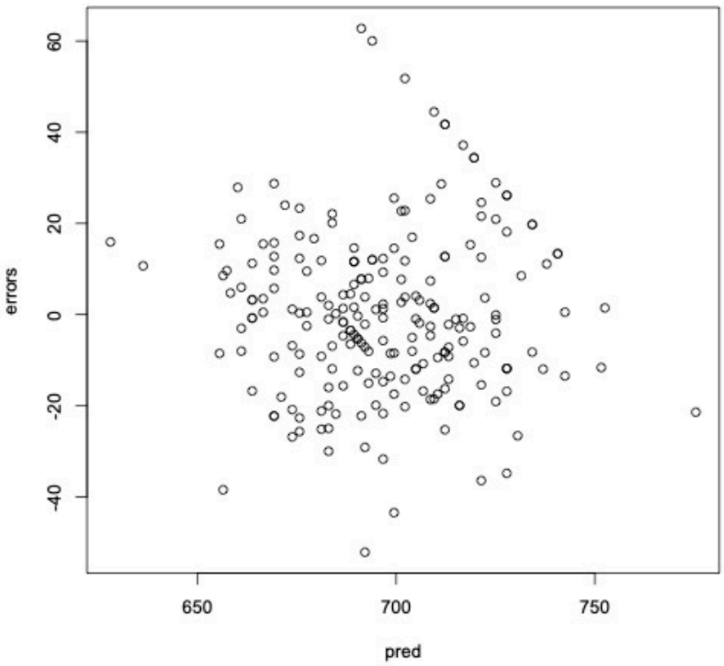
Question 2: Is there a significant relationship between prior standardized test scores in middle school math (6th, 7th, or 8th grade) and Algebra 1 End of Course Exam scores?

A linear regression analysis was used to examine the existence of a significant relationship between junior high math standardized test scores and Algebra 1 EOC exam scores. Before conducting the analysis, several assumptions were tested to determine if the data was fit for this analysis. The diagonal values of the hat matrix were also examined. The range of these values is (0.005, 0.06). This range exceeds the threshold of 0.02, indicating that there are outlier values influencing the model. The normality assumption was examined by analyzing a QQ plot. This plot is shown below in Graph 3. The equal variance assumption was examined using a plot of the residual values over the fitted values. This plot is shown below in Graph 4.

Graph 3. Linear Regression Normal QQ Plot



Graph 4. Residual Values vs. Predicted Values



Upon examination of the QQ plot, there appear to be some extreme values at the higher end of the scale. However, the majority of values fall on or near the line, indicating that the data being used comes from a mostly normal distribution. When examining the residual over predicted values plot, it is revealed that the values are spread out and do not follow a pattern. This indicates that the equal variances assumption was not violated.

A linear regression model was created to determine if there is a significant relationship between junior high standardized math test scores and Algebra 1 EOC exam scores. The adjusted R^2 = 0.619. This indicates that 61.9% of the variation in Algebra 1 EOC exam scores can be explained by junior high standardized math test scores. Table 9 below shows the estimated regression coefficients, standard errors, 95% confidence intervals (CI), and descriptive statistics for this linear regression model. The ANOVA table for the linear regression model is shown below in Table 10.

Table 9: Descriptive Statistics for Linear Regression Model

Dependent Variable	Independent Variable	Estimated Regression Coefficient	Standard Error	t-value	p-value	95% CI
Alg_score	**jr_high_score**	0.92	0.05	18.62	p < .001	(0.82, 1.01)

Table 10: ANOVA Table for Linear Regression Model

	Df	SS	MS	F value	p value
jr_high_score	1	111681	111681	346.66	p < .001
Residuals	211	67976	322	----	----

After analyzing the linear regression model, it can be concluded that there is a statistically significant relationship between junior high standardized math scores and Algebra 1 EOC exam scores; $F(1, 211) = 346.66$ and $p < .05$. This indicates that students with lower scores on junior high standardized math tests tend to have lower scores on the Algebra 1 EOC exam.

Summary

This chapter covered the results obtained from conducting a multiple regression analysis and a linear regression analysis with a data set. The data set contained information collected from students at a single school district. This information included the number of absences in 8th grade, the number of absences in 9th grade, free lunch status, junior high standardized math test scores, and Algebra 1 EOC exam scores.

The multiple regression analysis was conducted to determine if the number of absences in 8th grade, the number of absences in 9th grade, or free lunch status are significant predictors of Algebra 1 EOC exam scores. After conducting the multiple regression analysis, it was determined that 9th-grade absences were a significant predictor of Algebra 1 EOC exam scores. This shows that the number of absences a student has in 8th grade and a student's socioeconomic status are not statistically significant predictors of their Algebra 1 EOC exam score.

The linear regression analysis was conducted to determine if there is a significant relationship between junior high standardized math test scores and Algebra 1 EOC exam scores. After conducting the linear regression analysis, it was determined that there is a significant relationship between junior high standardized math test scores and Algebra 1 EOC exam scores.

Based on both analyses conducted using the data set, it can be concluded that junior high math test scores and attendance in 9th grade are the most statistically significant factors in predicting Algebra 1 EOC exam scores. This may indicate that school administrators and teachers should look more closely at these areas of data when determining if a student is likely to need additional instruction to earn an adequate score on the Algebra 1 EOC exam.

Chapter 5

Introduction

This chapter will summarize the research conducted for this study. Chapter 5 will relate the study's results to all other research aspects. This chapter will discuss the results of the research in relation to the motivation for completing this study. The results of the study will also be discussed in relation to the theoretical framework stated in Chapters 1 and 2. The results of this study will be compared to the results of similar studies referenced in Chapter 2. Finally, this chapter will discuss the limitations and possible improvements to this study.

Motivation

This study was conducted to find significant predictors for Algebra 1 EOC exams. In early 2020, schools in Ohio experienced a complete shutdown which necessitated online learning. In late 2020, many students returned to school using a hybrid model, and CDC requirements for quarantining caused students to miss many days of school during the 2020 - 2021 school year. After these events, scores for standardized testing plummeted for many school districts across the state. These scores were especially low in mathematics.

Algebra 1 is a course that is required for students in Ohio to graduate high school. A part of this requirement includes receiving a scaled score of 684 or higher on the Algebra 1 EOC exam given by the state of Ohio every spring. As was mentioned previously, scores had fallen significantly. This included Algebra 1 EOC exam scores. These falling scores were the main motivator for this study. The researcher wanted to determine if 8th-grade attendance, 9th-grade attendance, previous math test scores, and socioeconomic status are significant predictors of Algebra 1 EOC exam scores. The results of this analysis would then be used to help teachers and administration determine if students needed to be placed in remedial math courses alongside their required courses to give them a greater chance of attaining a score of 684.

The results of the multiple regression analysis revealed that previous math scores from standardized tests and 9th-grade attendance were statistically significant predictors of Algebra 1 EOC exam scores. Teachers and administrators should be advised to examine which students had less than proficient scores on 8th-grade standardized tests in math to help place students in remedial classes at the beginning of the 9th-grade school year. It is also recommended that administrators record attendance for the first quarter of the school year. This would allow the school to determine which students are exhibiting a pattern of chronic absence. These students should also be placed into remedial math courses. These recommendations are in line with the findings from the multiple regression analysis.

As the motivation for this study was to give students the chance to succeed at the Algebra 1 EOC exam on the first try, it may be beneficial for teachers and administrators to find ways to improve attendance in 9th grade. This could include finding a way to incentivize school attendance through the use of an extrinsic reward or by pairing students with counselors to determine the root causes of the attendance issue. Since attendance is a significant predictor of Algebra 1 EOC exam scores, it can be said that scores would improve if attendance is improved.

Theoretical Framework

This study is rooted in Maslow's Hierarchy of Needs Theory. This theory is used to explain human motivation. Maslow's Hierarchy is an eight-tiered model that shows human needs starting with the most basic, concrete needs and ascending to the more abstract needs. (McLeod, 2018). The first tier of the model is comprised of food, shelter, warmth, and other basic necessities to survive. The second tier is centered around safety and security. (McLeod, 2018).

Maslow theorized that people cannot meet the upper levels of the hierarchy without first meeting the lower levels. This means that people cannot have safety without first having shelter, for example. (McLeod, 2018). This pattern follows for the entirety of the hierarchy.

Maslow's theory can be applied to people in many different environments and situations around the world. One such environment is public schools. Students in public schools have a difficult time focusing on learning new information if they are not having their basic needs met. (Burleson & Thoron, 2014). These basic needs should be met at home. However, many are not, especially in areas with high levels of poverty. Schools can help students meet some of their needs by providing food and a safe place for the day, but schools cannot meet every basic need of the students. Although schools cannot meet every need of students, having a few of their needs met can alleviate some of the pressure and allow students to focus more energy on learning and acquiring new skills. (Burleson & Thoron, 2014).

The results of this study align with Maslow's theory. When conducting the multiple regression analysis, it was found that 9th-grade attendance was a significant predictor of Algebra 1 EOC exam scores. However, socioeconomic status (SES) was not found to be a significant predictor of Algebra 1 EOC exam scores. This was then investigated further. By further analyzing the data, it was found that SES is a significant predictor of 9th-grade attendance. This shows that students with low SES are predicted to miss more days of school than students with middle or high SES. This shows that students with low SES are more likely to attain a passing score on the Algebra 1 EOC exam because they are missing more days of school than their higher SES peers. This circles back to students who are not having their needs met at home. Students who are struggling with a lack of basic needs being met, a lack of safety, a lack of self-esteem, etc., are less likely to attend school. These students are not focused on building relationships and gaining knowledge. They are focused on meeting those unmet needs. (Burleson & Thoron, 2014). These students fit well within the confines of Maslow's theory. If they do not have adequate food and shelter and do not have a safe and secure environment, they are less likely to attend school and reach for the higher levels of the hierarchy.

Comparison with Similar Literature

When reviewing literature for this study, the topics the researcher focused on were the connection between poverty and chronic absenteeism, the connection between chronic absenteeism and academic achievement, and the compounding effects of chronic absenteeism across multiple years of school.

When researching the connection between poverty and chronic absenteeism, it was found that students from low SES backgrounds are likely to miss more school than their middle to high SES peers. (Balfanz & Byrnes, 2012). Students who live in poverty are more likely to miss school due to illness and injury of themselves or of a family member, as these students may be needed to help the sick or injured family member, or they may not be able to seek medical care. (Jensen, 2009). These same students are also more likely to develop depression and other mental illnesses due to the residual effects of living in poverty during their developmental years. (Jensen, 2009). Along with health issues, students living in poverty are also at risk to become transient. This is when the students and caregivers cannot find consistent housing for any number of reasons. This leads to the family moving around to different homes, or possibly not living in a home for periods of time. (Jensen, 2009). All of these reasons and many others make it more difficult for low SES students to attend school on a regular basis. In this study, it was found that SES is a significant predictor of 9th-grade attendance. More specifically, it was determined that students with low SES are predicted to miss 6.81 days of school more than middle and high SES peers each year. This shows that the analysis conducted in this study is consistent with the results found in similar studies.

Since 9th-grade attendance was a significant predictor of Algebra 1 EOC exam scores, the above findings show that students with low SES may be at a higher risk for achieving a lower score than students with middle or high SES. It certainly shows that students with more days absent during their 9th-grade year are more likely to achieve a lower score on the Algebra 1 EOC exam. This finding is consistent with previously conducted research. Parke and Kanyongo (2012) studied the effects of attendance and transience on students living in an urban school district in Pittsburgh, Pennsylvania. Their research found that students with higher numbers of days missed each year achieved a lower score on the math standardized tests given by the state of Pennsylvania. Parke and Kanyongo (2012) also found that when specifically looking at students in high school there is a significant relationship between attendance and standardized test scores in mathematics.

When students are absent from school across multiple years, the effects can often compound to create larger learning gaps, and causing students to fall significantly behind their peers in academica achievement. (Balfanz & Byrnes, 2012). When these learning gaps are created in the middle grades, (6th through 8th grade), students' high school graduation is often affected. This happens because students do not attend school often enough to recover from the learning gaps created by their absence. (Balfanz & Byrnes, 2012).

When the multiple regression analysis and linear regression analysis were conducted for this study, it was found that previous standardized test scores in math were significant predictors of Algebra 1 EOC exam scores. The scores used were from the most recent standardized test the student had taken for mathematics. These tests were from either 6th, 7th, or 8th grade. Since these standardized tests are given at the end of the school year, they are used to measure academic growth for each student from one year to the next. It can be said that students who have lower scores on standardized tests in middle school are showing learning gaps that have developed during that academic year or possibly prior to that year. Since this study showed that 61.9% of the variation in Algebra 1 EOC exam scores is explained by the regression on middle school math test scores. This shows that the majority of the Algebra 1 EOC exam scores are explained by previous learning. Students who have developed learning gaps in previous years are more likely to have lower scores on the Algebra EOC exam when they reach high school. These findings are consistent with the findings from Balfanz & Byrnes (2012) and Jensen (2009). Both of these pieces of literature assert that learning gaps created in early grades set children up for lower levels of academic achievement in high school.

Limitations and Generalizability

The limitations of this study are mostly connected to the way standardized testing is conducted. The first limitation of the study is connected to students' performance on the standardized tests. When conducting the analyses for this study, the researcher had to assume that all students had tried their best on each of these standardized tests. However, this may not have been the case for all students. There is no way for the researcher to have taken this into account when conducting analyses.

The second limitation to this study is related to standardized testing during the Covid-19 pandemic. During the 2019-2020 school year, schools were closed to protect from the unknown virus. This school closure across the state led to a cancellation of testing during that year. This cancellation and lack of testing made it difficult to use the same standardized test for all students. This led the researcher to use the most recent standardized test that students had taken prior to completing Algebra 1 and the Algebra 1 EOC exam. This meant that some cases included a score from 6th-grade, some from 7th-grade, and some from 8th-grade. Ideally, the scores used would have all been from the students' 8th-grade year to measure their skills immediately prior to entering 9th-grade and Algebra 1.

This study would be difficult to generalize to the greater population of students in the state of Ohio and even more so across the United States. This study is difficult to generalize because the sample of students was collected from one school district in a rural area of Ohio. The overall population of the district is not diverse culturally or socioeconomically. This lack of diversity makes it difficult to apply the results to groups of students in different areas of the state where more diversity may exist.

Improvements and Implications for Future Research

If this study were to be replicated, there are a few changes that could improve the outcome of the study. The first change to be suggested would be to use test scores from multiple school districts in the same area to gain a larger sample size. This would better represent the students from this area of Ohio and give greater insight into the effects of the variables included in this study. Another change to this study concerns the test scores from middle school. It would be more beneficial to use scores from tests taken immediately prior to students beginning Algebra 1. To do this, it is suggested that the researcher obtain scores from multiple districts and only use the most recent freshmen and sophomore classes, as those students would likely have an 8th-grade math test score and an Algebra 1 EOC exam score. This would allow the researcher to assess the scores from the same year for all students. This would limit the variability in skills needed to earn a proficient score on the test.

The results from this study open the door for future research to gain a broader understanding of which factors affect student test scores in Algebra 1. To further the research conducted in this study, future researchers may want to use student data from rural, urban, and suburban areas to determine if the same predictors are significant for students living and attending school in each of these environments. The cross comparison of students from these different environments may lead to revelations about the factors affecting student learning in different environments.

Another future research idea related to this study would use the comparison of students prior to the Covid-19 closings and after the Covid-19 closings. A researcher could use student testing data from school years that would have been unaffected by the Covid-19 school closings, the 2018-2019 school year for example. Then, the researcher could use student test data from years where students' middle grades math skills would have been impacted by the Covid-19 school closings. This may include students who were in 7th or 8th grade during the 2019-2020 school year or the 2020-2021 school year. The researcher could then compare the data and determine if there is a significant change in which predictors affect students' Algebra 1 EOC exam scores.

Conclusion

This study focused on determining which factors are significant predictors of a student scores on the Algebra 1 EOC exam and using the results to target groups of students to receive remediation in mathematics during their freshman year of high school. The results of this study are consistent with the results from several similar studies. Upon completing this study, it has been determined that attendance and prior math scores are the most significant predictors in determining if a student will achieve a high score on the Algebra 1 EOC exam. Furthermore, it was then determined that attendance can be linked to SES. This information leads the researcher to suggest that school administration find ways to help students attend school more often and take a closer look at middle school test scores when scheduling students for classes. These steps will help students achieve at a higher level and possibly navigate through some of the negative effects that poverty can place on their education and academic achievement.

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APPENDIX A

Appendix A contains proof of IRB approval for this study.

CERTIFICATE

OF COMPLETION

PHRP Online Training, Inc. certifies that

Mirranda Glasgow

has successfully completed the web-based course "Protecting Human Research Participants Online Training."

Date Completed: **2022-08-27** Certification Number: **2982201**

PHRP
Protecting Human
Research Participants
Online Training

Shawnee State University

Study # 2022-13

Exempt Review Application

Title of Research Project:
Factors Related to Algebra 1 Students' Success on End of Course Exams

Name of Principal Investigator	Email Address	Phone Number
*Douglas Darbro	ddarbro@shawnee.edu	740-351-3341

Department(s)/Division/Agency: Department of Mathematics

Name(s) of Co-Investigators:	Email address:	Faculty	Student	Other
Mirranda Glasgow	glasgowm@mymail.shawnee.edu	☐	☒	☐
		☐	☐	☐
		☐	☐	☐
		☐	☐	☐
		☐	☐	☐
		☐	☐	☐

*Please place an asterisk by the investigator name(s) whose NIH certificate(s) is/are already on file with the IRB, if the certificate is less than 3 years old.

Please place a check mark next to the category that best describes your research. You may check more than one category.

- ☐ Research conducted in established or commonly accepted educational settings, involving normal educational practices, such as (a) research on regular and special education instructional strategies, or (b) research on the effectiveness of or the comparison among instructional techniques, curricula, or classroom management methods.
- ☒ Research involving the use of educational tests (e.g., cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures or observation of public behavior, unless: (a) data obtained is recorded in such a manner that human subjects can be identified, directly or through identifiers linked to the subjects; and (b) any disclosure of the human subjects' responses outside the research could reasonably place the participants at risk of criminal or civil liability or be damaging to the participants' financial standing, employability, or reputation. No videotaping or photography is allowed for data collection. You may not collect data from appointed public officials or candidate for public office.

☒ Research involving the collection or study of existing information, documents, records, pathological specimens, or diagnostic specimens, if these sources are publicly available or if the information is recorded by the investigator in such a manner that subjects cannot be identified, directly or through identifiers linked to the subjects.

☐ Research and demonstration projects that are conducted by or subject to the approval of supporting agencies, and which are designed to study, evaluate, or otherwise examine: (a) public benefit or service programs; (b) procedures for obtaining benefits or services under those programs; (c) possible changes in or alternatives to those programs or procedures; or (d) possible changes in methods or levels of payment for benefits or services under those programs.

☐ Taste and food quality evaluation and consumer acceptance studies, (a) if wholesome foods without additives are consumed or (b) if a food is consumed that contains a food ingredient at or below the level, and for a use, found to be safe, or agricultural chemical or environmental contaminant at or below the level found to be safe, by the Food and Drug Administration and approved by the Environmental Protection Agency or the Food Safety and Inspection Service of the U.S. Department of Agriculture.

Does your research include at least one of the above criteria? Yes X **No**

1. Describe the key demographics (age, SES, ethnicity, geographic locations, gender, etc) of the sample that you wish to obtain.

The participants will be current students at Buckeye Local high school. The students will be aged 15 - 17, current sophomores and juniors. There will be no exclusion criteria for ethnicity or gender.

1a. What is the greatest number of participants that will be recruited? 300

1b. How will participants be recruited

I will be using data that has already been collected and recorded by the district.

2. Will participants be remunerated for their participation? Yes No X

2a. If so, how will participants be remunerated? Please indicate the type of remuneration and the amount. For instance, the participants will be given a $10 Amazon Gift Card for participation or the participants will receive 3% of their final grade in extra credit in their Introduction course.

N/A

2b. If participants do not complete the study, will partial or full remuneration be given? Please describe how that will be determined.
N/A

3. What direct benefits (other than remuneration) exist for the participants who participate?
none

4. What direct risks could the participants potentially face? Check all that apply.

_____ Risk of breach of confidentiality or privacy

_____ Risk of coercion by researcher(s)

_____ Risk of psychological harm

_____ Risk of physical harm

_____ Other potential risk: ____

If you checked any direct risks in Item 4, then you should complete the "Expedited and Full Review Application."

5. Will the participants be informed of the risks and benefits of the study? Yes No X

5a. If so, how will the participants be informed?
N/A

5b. Please check each box if the following criteria match your research.

☒ The research involves no greater than minimal risk.
☒ It is not practicable to conduct the research without a waiver of informed consent or alteration to informed consent.
☒ Waiving or altering the informed consent will not adversely affect the subjects' rights and welfare.
☒ The consent document would be the only record linking the subject and the research, and the principal risk would come from a breach of confidentiality.

5c. Do you wish to waive the signed informed consent? Yes X No

SSU IRB Approved TH 11/14/2022 | 2:08 P

Shawnee State University

Study # 2022-13

In submitting this form and the corresponding documents, I acknowledge that I have completed Human Research Participants training and that I understand and will uphold the rights of human participants. I also verify that all information contained in this form and any other corresponding documentation is correct based on my knowledge. I understand that I may not have contact with any research participants until the Shawnee State University IRB has given me their approval. I also understand that I must file an ***Amendment/Modification Form*** if my project extends beyond a year from my approval date and I must file a ***Final Study Form*** with all consent forms once the study is complete.

DocuSigned by: Douglas Darbro
Signature of Principal Investigator 1

DocuSigned by: Mirranda Glasgow
Signature of Co-Investigator 2

Signature of Co-Investigator 3

Signature of Co-Investigator 4

Signature of Co-Investigator 5

Signature of Co-Investigator 6

Date of Submission: 10/25/2022 | 3:45 PM EDT

Please compile attachments into one document for each category. If any forms below are not applicable, please attach reasons why.

Human Research Training Certificates:

Data Collection Questions and Forms:

Research Summary:

Consent Forms:

Assent Forms:

Advertisements:

Revisions Requested Yes No X

IRB Chair Signature DocuSigned by: Tim Hamilton

DS DDD

Date sent for revision (if applicable): ____________

Please attach revisions requested with changes clearly marked

Changes marked

Final copy

Rev. 9/3/2013;1/24/22 4

I. Summary of Research Questions:

Research has shown that students who are not having their needs fully met at home are at an increased risk for not succeeding academically at school. This risk is often compounded by students having excessive absences at school. Studies typically look at a student's socio-economic status (SES) and attendance record over the course of an academic year, then look at test scores or grades at the end of that year. In my study, I will look at student data from the previous academic year. I want to determine if students' socio-economic status and attendance during their 8th grade year affects their performance on the Algebra 1 End of Course (EOC) Exam taken in their 9th grade year. This study seeks to determine factors that may indicate success or non-success on the (EOC) Exams prior to enrollment in Algebra 1.

Primary research Question: Are 8th grade attendance and socio-economic status statistically significant predictors of end of course exam scores in 9th grade?

II. Methodology, Research Design, and Procedures

The research design will include students who are currently sophomores and juniors (10th and 11th grade) at Buckeye Local High School (BLHS). As a public high school, we keep records of students' attendance, free lunch status, and standardized test scores. I will collect data on students' 8th grade attendance, free lunch status, 8th grade scores on the math standardized test, and their previously taken Algebra 1 (EOC) exams from databases already accessible to myself and my school administrators. Data will be cleansed of identifying factors before analysis.

III. Data Analysis and Reporting

Once the data is collected, I will conduct an ANOVA to determine if there is a difference in EOC Exam scores based on 8th grade math test scores. I will also conduct a linear regression analysis to determine if attendance, SES, or both of these factors predict success on the Algebra 1 EOC Exam taken in 9th grade.

The results of this study will be included in a paper to fulfill the requirements for the Master's of Science in Mathematics program at Shawnee State University. The results may likely be published. I may also share the results of this study with administrators at BLHS to better help future students.

There is no risk of breaching confidentiality since the data will be cleansed of identifying information before analysis.

IV. Timeline

Data collection will begin once approval is received from the IRB at Shawnee State University. All data will be collected by May 31, 2023.

I will not be using any type of survey or form to collect the data. It has already been recorded by the school district.

I have chosen to waive the consent and assent forms since the data I will be using has already been recorded by the district, and it is data that all teachers of our district already have access to.

I have chosen to waive the consent and assent forms since the data I will be using has already been recorded by the district, and it is data that all teachers of our district already have access to.

I will not be using any advertisements to recruit participants.

APPENDIX B

The data set used to conduct the study

subject	9thAbs	8thAbs	freelunch	algscore	jrhighscore
1	17.01	0	0	704	721
2	29.93	7.68	0	711	742
3	23.18	8.04	0	696	729
4	11.55	6.41	0	724	739
5	9.47	5.12	1	740	724
6	19.53	7.51	0	706	739
7	13.08	5.91	0	685	735
8	11.22	7	1	706	705
9	17.21	17.84	0	704	718
10	0	4.91	0	709	717
11	32.05	1.63	0	746	742
12	25.65	28.22	1	734	732
13	10.41	6.3	0	709	713
14	19.56	3.78	0	716	721
15	4.83	1.38	0	749	753
16	11.88	0.97	0	706	735
17	24.29	12.19	1	709	708
18	17.35	10.5	0	709	718
19	3	3	0	711	721
20	15.1	11	0	740	746
21	11.15	3.15	0	746	735
22	27.15	10.25	0	724	713
23	11.57	12.04	0	716	742
24	21.51	9.17	0	729	758
25	84.3	17.88	0	746	739
26	21.13	11.41	0	740	768
27	17.87	9.78	0	734	721
28	16.16	18.48	0	699	726
29	19.09	8.52	0	716	732
30	31.51	17.35	0	726	749
31	0	7.08	0	743	735
32	8.86	5.45	0	734	735
33	14.22	11.03	0	704	726
34	27.67	4.99	0	685	695

35	7.53	4.14	0	711	726
36	7.45	5.18	0	743	758
37	13.37	20.72	0	704	713
38	4.99	2.76	0	706	726
39	25.38	0	1	656	711
40	1.69	1.75	0	699	702
41	5.64	11.79	0	754	702
42	7.12	14.41	1	711	722
43	28.6	17.55	1	682	714
44	15.22	7.78	0	691	697
45	17.54	5	1	721	739
46	3.38	9.47	1	754	725
47	8.24	4	0	693	685
48	4.31	5	0	714	714
49	34.85	13	0	644	633
50	8	6.8	1	701	700
51	15.05	0.55	0	696	700
52	15.12	10.14	0	685	702
53	8.08	2.28	0	754	756
54	5.29	10.43	0	711	730
55	12.27	13.68	1	725	739
56	20.17	10	1	706	705
57	22.23	6.49	0	650	685
58	13.65	8.45	1	754	742
59	15.15	6.46	1	725	725
60	10.51	9.04	0	665	664
61	31.6	12.21	1	698	708
62	0.25	1.09	0	665	708
63	10.5	8.52	0	726	736
64	32.05	23.9	0	696	716
65	10.9	6.39	1	754	725
66	10.19	11.02	0	754	739
67	14	12.04	1	685	700
68	30.28	24.46	0	676	685
69	15.38	20.17	1	709	733
70	14.25	21.12	0	696	708
71	11.04	7.47	0	688	697
72	11.35	14.5	1	704	725
73	5.21	1.41	1	682	675
74	31.01	7.44	1	716	742

75	9.46	3.22	0	682	678
76	6.55	9.01	1	725	711
77	11.14	11.36	1	701	700
78	24	23.16	0	725	714
79	37.28	18.39	1	691	711
80	40.29	24.48	1	754	742
81	20.68	3.13	1	670	675
82	17.99	4.65	0	711	722
83	31.91	16.44	1	691	700
84	10.54	6.21	0	660	678
85	12.49	6.7	0	696	681
86	3	2.38	0	688	668
87	13.7	9.97	0	706	694
88	11.36	12.27	0	693	691
89	10.4	5.5	0	754	705
90	6.11	3.94	0	656	691
91	13.05	4	1	679	678
92	3.61	10.23	0	696	725
93	10.31	3.73	0	754	749
94	9.33	17.32	1	725	752
95	6	1.15	0	714	711
96	15.7	42.03	0	699	716
97	5.14	11.26	1	669	702
98	19.82	7.53	0	688	714
99	9	4.04	1	688	685
100	20.67	5.06	0	701	700
101	14.49	11.42	1	704	694
102	11.18	4.21	0	754	756
103	28.51	19.09	1	691	722
104	6.39	0.37	0	716	730
105	20.16	5.63	0	754	733
106	11.62	9.83	0	704	725
107	11.18	8.04	1	677	694
108	5.78	2.87	1	699	708
109	11.18	8.68	0	672	694
110	19.65	5.39	0	660	691
111	15.71	13.43	0	754	733
112	32.08	14.6	1	754	714
113	10.79	7.17	0	754	769
114	18.49	7.12	1	754	749

115	6	8.3	1	696	719
116	1	1	0	618	664
117	19.06	4.77	0	714	728
118	15.4	7.38	1	699	702
119	8	4	0	754	794
120	14.66	9	0	714	736
121	11	9.85	0	706	708
122	1.07	4.09	1	725	725
123	31.42	18.5	1	704	700
124	23.87	9	0	754	730
125	16.88	19.19	1	706	714
126	34.53	14.03	1	691	708
127	29.19	8.12	1	704	745
128	17	16.45	0	672	691
129	24.67	23.3	1	721	716
130	22.48	15.65	0	699	685
131	3.38	2	1	754	722
132	9.98	6.69	1	685	678
133	26	15.1	0	667	683
134	15.23	13.9	1	682	693
135	60.49	52.06	1	667	669
136	4.31	0.41	1	685	693
137	39.32	35.78	1	663	666
138	5.47	10.37	0	696	703
139	19	3.73	1	696	706
140	17.73	9.54	1	701	704
141	9.52	7.96	1	690	721
142	48.46	36.81	1	687	687
143	15.57	1.05	0	667	665
144	47.19	29.83	0	690	710
145	32.35	14.39	0	678	687
146	16.71	11.57	1	653	685
147	29.24	32.3	1	675	672
148	7.39	2.65	0	693	717
149	27.27	12.74	0	716	742
150	14.92	15.09	1	701	723
151	1	49.73	0	682	669
152	20.22	12.88	1	647	683
153	16.11	13.33	1	667	672
154	24.02	10.03	1	696	729

155	9.51	2	0	682	706
156	16.92	11.2	0	687	725
157	15.49	4.58	0	685	710
158	35.81	23.58	0	663	703
159	20.67	7.84	1	682	697
160	22.54	8.15	0	704	717
161	28.97	20.43	1	667	693
162	99.86	22.36	1	685	699
163	60.17	45.08	0	647	678
164	35.67	28.79	1	667	675
165	19.46	4.74	1	675	706
166	24.31	16.54	0	675	678
167	14.31	4.88	0	640	703
168	23.81	9.23	1	663	695
169	31.25	8.74	0	682	708
170	27.49	9.22	1	685	697
171	23.66	15.2	1	653	693
172	29.33	9.79	0	663	693
173	35.51	13.88	1	647	672
174	17.05	13.81	1	682	711
175	59.61	16.54	1	663	685
176	32.35	23.64	1	658	693
177	32.63	12.73	0	667	672
178	21.41	28.51	0	685	703
179	26.77	7.75	0	678	704
180	35	12.72	0	667	685
181	94	19.92	1	675	708
182	60.14	27.74	0	653	683
183	11.43	16.87	1	685	701
184	40.16	16.88	1	653	669
185	40.29	28.64	0	663	672
186	42.06	25.3	0	693	699
187	15	6.08	1	690	701
188	72.74	28.92	1	671	663
189	33.69	13	1	706	721
190	9.66	7.15	1	678	701
191	83.68	65.87	1	647	642
192	4.4	5.66	0	693	717
193	17.74	11.46	0	690	719
194	19.83	12.81	0	682	699

195	47.55	60.45	1	685	697
196	6.92	5.33	0	685	704
197	23.98	21.31	0	647	663
198	13.71	10	1	647	678
199	4.68	18.33	1	693	723
200	46.83	20.06	1	696	689
201	22.11	7.43	0	693	742
202	34.31	3.6	1	675	687
203	69.9	26.27	1	653	680
204	19.15	14.24	0	663	672
205	38.2	37.07	0	685	691
206	31.96	31.11	1	698	678
207	31.57	86.82	1	658	669
208	27.33	12.99	1	685	701
209	16.14	12.18	1	675	683
210	10.69	1	1	671	697
211	21.33	18.57	0	713	729
212	14.93	12.89	1	685	699
213	36.73	15.5	1	690	703

BIBLIOGRAPHY

Mirranda A. Glasgow

Candidate for the Degree of

Master of Science Mathematics

Thesis: **FACTORS CONTRIBUTING TO STUDENT SUCCESS ON ALGEBRA 1 END-OF-COURSE EXAMS**

Major Field: Mathematics

Biographical: 4th-grade mathematics teacher. Hoping to teach CCP courses for my high school or begin teaching at the college level

Personal Data: I began teaching Algebra 1 in 2019 and taught it for four years. I am now transitioning to teach 4th grade for the upcoming school year. My husband and I enjoy traveling and spending time outdoors during the summer.

Education: B. S. in Middle Childhood Education, Ohio University, Athens, Ohio

Completed the requirements for the Master of Science in Mathematics, Portsmouth, Ohio in August 2023.

7/25/2023

ADVISER'S APPROVAL: Type Adviser's Name Here

Dr. Douglas Darbro

1